

CSS Flexbox & Card Components

Build beautiful layouts like Zomato, Flipkart & BYJU'S!

```
.container { display: flex; }  
  
flex-direction: row | gap: 16px;  
  
border-radius: 12px;
```

What We're Building Today

By the end of this session, you'll create card layouts like the ones on your favourite apps!

01 What is CSS?

Colors, fonts, sizes & layout language

02 What is Flexbox?

Container vs Item — the tiffin box model

03 Flexbox Properties

flex-direction, justify-content, align-items, gap

04 Card Components

Image • Title • Description • Button zones

05 Build a Card — Live!

5-step walkthrough: HTML → CSS → Layout

What is CSS?

✨ Analogy

HTML is the skeleton of a webpage. CSS is the decoration — like adding colourful embroidery to a plain kurta to make a beautiful festival outfit!

Without CSS

Messy stacked text

No colours

Hard to read

With CSS

Clean card layout

Rich colours & media

Intuitive & beautiful

CSS Syntax

```
.card {  
  background-color: white;  
  border-radius: 12px;  
  padding: 16px;  
}
```

What is Flexbox?

Tiffin Box Analogy

The tiffin dabba = Flex Container (holds everything).
Each compartment (rice, dal, sabzi) = Flex Item — neatly organised inside, automatically fitting perfectly!

Element	Role	Real-Life
Flex Container	PARENT — gives instructions	Teacher in assembly
Flex Item	CHILD — follows instructions	Students obeying teacher

Activating Flexbox — the ON Switch

```
.container {  
  display: flex;    /* Turns ON Flexbox! */  
}
```

 *display: flex is applied to the CONTAINER, not to the items. The container becomes the 'boss'!*

SECTION 3

Flexbox Properties

flex-direction



row

LEFT → RIGHT (default)



row-reverse

RIGHT → LEFT



column

TOP → BOTTOM



column-reverse

BOTTOM → TOP

justify-content – Left/Right Spacing



Thali analogy — where do you place the bowls?

flex-start

Items crowd LEFT

center

Items in the MIDDLE

flex-end

Items crowd RIGHT

space-between

Equal space between

space-evenly

Equal space everywhere

align-items & gap

align-items — Up/Down Alignment

📖 Bookshelf analogy — where do the books sit on the shelf?



flex-start



center



flex-end



stretch

Books on TOP shelf

Books at MID — balanced

Books on BOTTOM

Books fill full height (default)

gap & flex-wrap

gap: 16px

Adds breathing room between flex items — like gutters in a magazine grid. No more manual margins!

flex-wrap: wrap

When there's no room, cards move to the next line — like books overflowing to the next shelf!

Card Components

A card is a self-contained rectangular box displaying info beautifully — image + title + description + button.

Zomato

Photo + Name + Rating + Delivery time

Flipkart

Image + Name + Price + Add to Cart

BYJU'S

Image + Class + Teacher + Enroll button

Anatomy of a Card

Zone	HTML Element	Purpose
Image area	<code>div.card-image</code>	Visual that catches the eye first
Title	<code>h3</code>	Name or headline of the item
Description	<code>p</code>	Short supporting text
Button (optional)	<code>button</code>	Call to action — Order, Buy, Enroll

Build a Card — Steps 1 to 3

1 HTML Structure — The Card Skeleton

```
<div class="card">
  <div class="card-image"></div>
  <h3>Pizza Palace</h3>
  <p>Delivered in 30 min!</p>
</div>
```

2 Card Base CSS — Activate Flexbox

```
.card {
  display: flex; /* ON! */
  flex-direction: column;
  border-radius: 12px;
  box-shadow: 0 2px 8px ...
```

3 Image Placeholder — The Picture Frame

```
.card-image { width: 100%; height: 120px; background-color: #FFA500; }
```

width: 100% fills the full card width · height: 120px creates a fixed image zone · background-color = orange placeholder

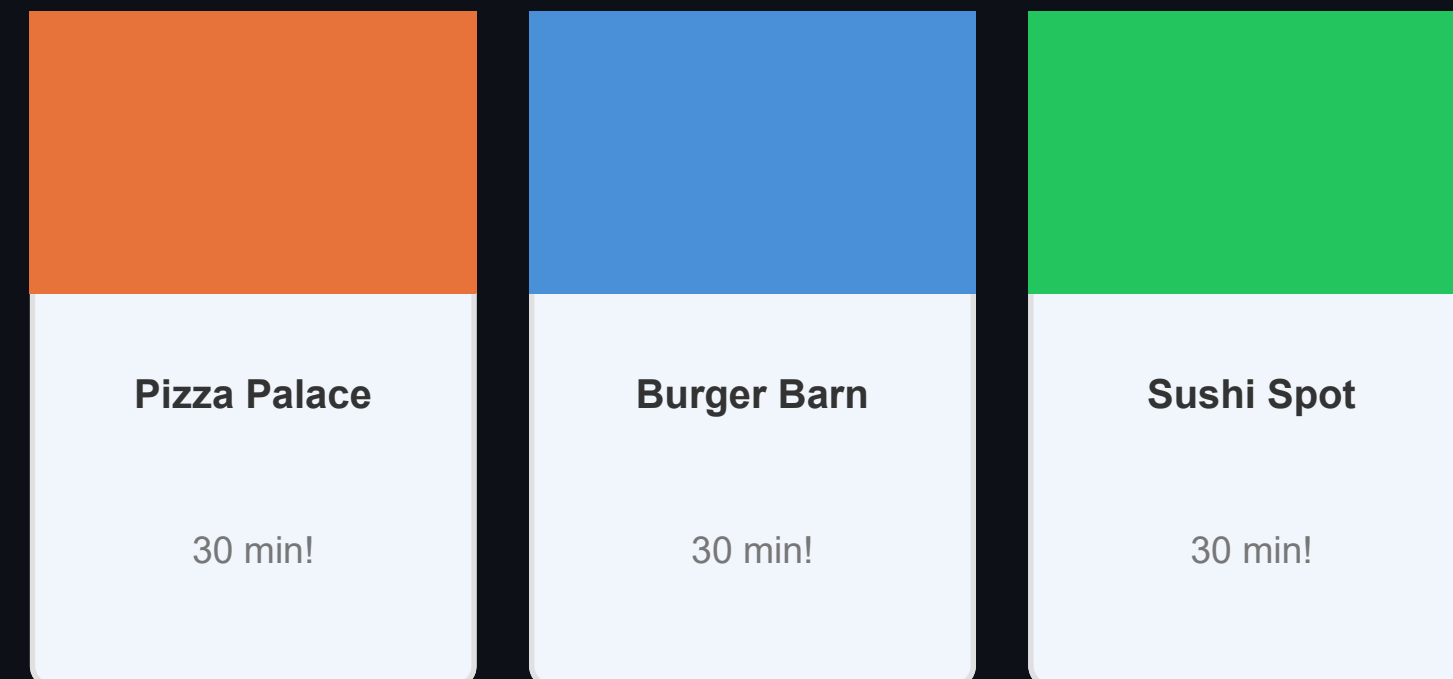
Build a Card — Steps 4 & 5

4 Text Styling — Title and Description

```
.card-title { font-size: 16px; color: #333; }  
.card-description { font-size: 13px; color: #777;  
  text-align: center; padding: 0 12px 12px; }
```

5 Multiple Cards in a Row — The Card Grid!

```
.card-row {  
  display: flex;  
  justify-content: space-between;  
  gap: 16px;  
  flex-wrap: wrap;  
}
```



Recap Flash Cards

The magic tools we learned today — one card, one concept!

display: flex

Turns Flexbox ON

💡 *The ON Switch!*

flex-direction: row

Items go left → right

💡 *Students in classroom seats*

flex-direction: column

Items go top → bottom

💡 *Floors stacked in a building*

justify-content: center

Centre items on main axis

💡 *Dal bowl at centre of thali*

justify-content: space-between

Equal space between items

💡 *Bowls spread on thali*

align-items: center

Centre items on cross axis

💡 *Books at mid-shelf*

gap: 16px

Space between flex items

💡 *Breathing room between cards*

border-radius: 12px

Rounded corners

💡 *Friendly greeting card look*

flex-wrap: wrap

Items wrap to next line

💡 *Books overflow to next shelf*

Homework & Next Session

1 Find 3 Real Cards

Open Zomato, Flipkart, or any app. Screenshot or sketch 3 card components in your notebook.

2 Card Element List

For each card you found, list what's inside — image, title, rating, price, button, etc.

3 Draw Your Own Card

Sketch a card for your favourite food, book, or game. Label each zone: image, title, description, button.

★ Bonus Challenge (Optional)

Add a hover effect to your card — cards lift up when you mouse over!

```
.card:hover { transform: translateY(-4px); box-shadow: 0 8px 20px ...; }
```

Coming Next: CSS hover effects, transitions & mobile-friendly layouts with media queries!

You Built Your First **Flexbox Layout!**

That's what real web developers do every day.

Keep practising, keep exploring, and you are on your way to becoming a coding superstar! 🌟

`display: flex`

`flex-direction`

`justify-content`

`align-items`

`gap`

`border-radius`